

SECURITY INCIDENT REPORT

Mini SOC — Active Directory Attack Detection

Environment: homelab.local | Date: June 3, 2026 | Classification: CONFIDENTIAL

SEVERITY

CRITICAL

STATUS

RESOLVED

MITRE

3

TECHNIQUES

1. Executive Summary

This report documents a simulated multi-stage cyberattack against the **homelab.local** Active Directory environment as part of a controlled home lab exercise. The attack was executed from Kali Linux (192.168.10.4) targeting a Windows Server 2022 Domain Controller (192.168.10.1) and Windows 10 workstation (192.168.10.2).

The Wazuh SIEM platform deployed on Ubuntu Server (192.168.10.3) successfully detected all attack phases and generated corresponding security alerts. The exercise validates detection capabilities against real-world Active Directory attack techniques mapped to MITRE ATT&CK.

2. Environment Overview

Role	Hostname	IP Address	OS / Software
SIEM Server	ubuntu-server	192.168.10.3	Ubuntu Server + Wazuh 4.10
Domain Controller	DC01	192.168.10.1	Windows Server 2022 + AD DS
Workstation	WIN10	192.168.10.2	Windows 10 + Wazuh Agent
Attacker	kali	192.168.10.4	Kali Linux + Impacket

Domain: homelab.local | **Users:** administrator, Admin, jan.kowalski, piotr.wisniewski, anna.nowak, sqlsvc

3. Attack Timeline

Time	Phase	Technique	MITRE ID	Description
08:00	Phase 1	SIEM Setup	–	Wazuh agent deployed, baseline logs verified
08:15	Phase 2	Password Spraying	T1110	Multiple failed logons from 192.168.10.4
08:30	Phase 3	PowerShell Abuse	T1059.001	Recon commands via PowerShell on WIN10
08:45	Phase 4	Atomic Red Team	T1059.001	Simulated adversary via Invoke-AtomicTest
09:00	Phase 5	Kerberoasting	T1558.003	RC4 ticket for sqlsvc obtained and cracked

4. Attack Details

4.1 Password Spraying — T1110

A password spraying attack was executed from Kali Linux against all domain accounts (Admin, jan.kowalski, piotr.wisniewski, anna.nowak) using a single common password. This technique avoids account lockout by limiting attempts per account.

Detection: Wazuh custom rule 100002 triggered — Multiple Failed Logon Attempts. Event ID 4625 observed in bulk from source IP 192.168.10.4.

4.2 PowerShell Abuse — T1059.001

PowerShell was used on WIN10 for internal reconnaissance:

- Get-Process
- Get-LocalUser
- Get-ComputerInfo

Detection: Custom Wazuh rule 100001 triggered on PowerShell command-line pattern. Event ID 4688 (Process Creation) and 4104 (Script Block Logging) recorded.

4.3 Atomic Red Team — T1059.001 / T1003

Atomic Red Team tests Invoke-AtomicTest T1059.001 and T1003 generated realistic Windows Security events without deploying actual malware.

Detection: Process creation events captured by Wazuh agent. MITRE ATT&CK mapping automatically applied.

4.4 Kerberoasting — T1558.003 [CRITICAL]

Impacket GetUserSPNs was used to request a Kerberos Service Ticket for sqlsvc (SPN: MSSQLSvc/dc01.homelab.local:1443). The RC4 TGS hash was extracted and cracked offline:

```
impacket-GetUserSPNs homelab.local/admin -dc-ip 192.168.10.1 -request
```

Result: \$krb5tgs\$23\$ hash obtained for sqlsvc — password successfully cracked.

Detection: Custom Wazuh rule 100003 triggered on Event ID 4769 with RC4 encryption type 0x17.

5. Indicators of Compromise (IOC)

Type	Value	Description
IP Address	192.168.10.4	Attacker machine (Kali Linux)
User Account	sqlsvc	Kerberoasted service account
User Account	admin	Account used to initiate Kerberoasting
SPN	MSSQLSvc/dc01.homelab.local:1443	Service Principal Name targeted
Hash Type	\$krb5tgs\$23\$	RC4 Kerberos TGS ticket hash — cracked
Event ID	4625	Failed logon — Password Spraying
Event ID	4769	Kerberos ticket request — Kerberoasting
Event ID	4688	Process creation — PowerShell execution

6. MITRE ATT&CK Mapping

Technique ID	Technique Name	Tactic	Severity	Detected
T1110	Brute Force / Password Spraying	Credential Access	HIGH	YES
T1059.001	PowerShell	Execution	MEDIUM	YES
T1003	OS Credential Dumping	Credential Access	HIGH	YES
T1558.003	Kerberoasting	Credential Access	CRITICAL	YES

7. Custom Wazuh Detection Rules

Three custom rules created in `/var/ossec/etc/rules/local_rules.xml`:

Rule ID	Level	Description	MITRE
100001	10	Suspicious PowerShell Execution Detected	T1059.001
100002	12	Multiple Failed Logon Attempts (5 in 60s)	T1110
100003	14	Kerberoasting — RC4 Ticket Requested	T1558.003

8. Recommendations

#	Priority	Recommendation	Addresses
1	CRITICAL	Enforce MFA for all privileged accounts	T1110, T1558.003
2	CRITICAL	Disable RC4 encryption in Kerberos policy (enforce AES)	T1558.003
3	HIGH	Implement Account Lockout Policy (5 failed attempts / 30 min)	T1110
4	HIGH	Enable PowerShell Script Block Logging (Event ID 4104)	T1059.001
5	HIGH	Remove unnecessary SPNs from standard user accounts	T1558.003
6	MEDIUM	Deploy Privileged Access Workstations (PAW)	T1059.001, T1003
7	MEDIUM	Enable PowerShell Constrained Language Mode	T1059.001
8	LOW	Quarterly AD audits with BloodHound / PingCastle	General

9. Conclusion

This exercise successfully validated the Mini SOC environment's capability to detect multi-stage Active Directory attacks. All attack phases were detected by Wazuh SIEM, including the most critical — Kerberoasting — which resulted in service account credential compromise.

The project demonstrates end-to-end SOC analyst skills: infrastructure setup, SIEM integration, threat simulation using MITRE ATT&CK techniques, custom rule development, dashboard creation, and formal incident reporting — directly aligned with SOC L1/L2 Analyst responsibilities.